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BIOCLASS S.5 SAMPLE ITEMS
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ITEM 3

In **Kiryandongo District**, a group of farmers started storing harvested cassava roots in tightly sealed plastic bags to reduce drying. After a few days, the cassava developed a **strong smell, soft texture, and discoloration**. Some farmers who consumed it reported **stomach discomfort and weakness**.
Further observations showed:

- Cassava stored in open baskets remained firm longer
- Bags kept in cooler areas had less spoilage
- Cut cassava pieces spoiled faster than whole ones
- Washing cassava before storage delayed spoilage
- Some farmers noticed liquid accumulation inside the bags

Table 3: Cellular and Biochemical Indicators in Cassava Tissues

Parameter	Fresh Cassava	Stored Cassava
Oxygen availability (%)	100	40
Rate of anaerobic respiration (%)	10	80
Enzyme activity (respiratory enzymes) (%)	100	65
Cell membrane integrity (%)	100	50
Accumulation of toxic metabolites (%)	5	70

Task

(a) Analyse how changes in **oxygen availability, respiration type, enzyme activity, membrane integrity, and metabolite accumulation** affected the quality of cassava and human health.

(b) Propose suitable measures that would reduce spoilage and maintain safe consumption.

(a) *Reduced* ***oxygen availability*** *(100% to 40%) limits aerobic respiration in cassava cells;* ***electron transport chain*** *activity decreases due to lack of final electron acceptor (**O_2**); reduced activity of* ***cytochrome c oxidase*** *lowers proton gradient formation and* ***ATP synthesis****.*
Increased ***anaerobic respiration*** *(10% to 80%) activates fermentation pathways;* ***pyruvate*** *is converted into toxic end-products (e.g., ethanol/organic acids); low ATP yield (2 ATP per glucose) fails to sustain cellular processes; accumulation of toxic metabolites causes tissue damage and unpleasant smell.*
Reduced ***enzyme activity*** *(100% to 65%) slows metabolic reactions; altered enzyme structure due to poor conditions reduces enzyme–substrate complex formation; decreases efficiency of respiration and other metabolic pathways.*
Loss of ***cell membrane integrity*** *(100% to 50%) disrupts the* ***phospholipid bilayer****; increased permeability leads to leakage of cell contents and entry of microorganisms; loss of compartmentalisation accelerates tissue breakdown and softening.*
High ***toxic metabolite accumulation*** *(5% to 70%) results from anaerobic metabolism and microbial activity; toxic substances damage cells and irritate human digestive tissues when consumed, causing discomfort and weakness.*
Cut surfaces increase surface area for microbial entry; trapped moisture promotes microbial growth; together accelerate cellular breakdown and spoilage.

(b) Strategies

Ventilated storage systems *increase oxygen availability; maintain aerobic respiration and efficient ATP production; reduce formation of toxic metabolites.*
Cool storage conditions *lower enzyme activity rate; reduce respiration rate and slow metabolic breakdown of tissues.*

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Avoid cutting before storage *limits exposed surfaces; reduces microbial invasion and preserves membrane integrity.*
Washing cassava before storage *removes surface microorganisms; reduces microbial respiration and toxin production.*
Use of perforated containers *allows gas exchange; prevents accumulation of CO_2 and moisture that favour anaerobic conditions.*
Early processing or consumption *limits time for metabolic deterioration; maintains tissue structure and prevents toxin build-up.*

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ITEM 4
In **Soroti District**, a fish farming project introduced high stocking densities in ponds to increase production. After some weeks, fish showed **reduced movement, slow growth, and increased mortality**. A field officer noted:

- Fish near water inlets appeared more active
- Ponds with aquatic plants had slightly better survival
- Feeding rates remained the same despite poor growth
- Some fish had pale body coloration
- Water appeared cloudy in heavily stocked ponds

Table 4: Cellular and Tissue Functional Indicators in Fish

Parameter	Normal Conditions	High Density Conditions
Dissolved oxygen (mg/L)	8.0	3.5
Mitochondrial ATP production (%)	100	50
Muscle tissue activity (%)	100	55
Enzyme activity (respiratory enzymes) (%)	100	60
Gill surface efficiency (%)	100	65

Task
(a) Evaluate how changes in **oxygen levels, ATP production, enzyme activity, tissue function, and gill efficiency** affected fish survival and growth.
(b) Suggest practical improvements that would increase fish productivity and reduce mortality.

RESPONSE
(a) Evaluation
*Reduced **dissolved oxygen** (8.0 to 3.5 mg/L) limits oxygen diffusion across gill surfaces; decreased availability of oxygen for **aerobic respiration** reduces efficiency of the **electron transport chain**.*
*Reduced **mitochondrial ATP production** (100% to 50%) results from limited oxygen supply; decreased activity of **ATP synthase** reduces energy yield; insufficient ATP affects active processes such as muscle contraction and ion transport.*
*Lower **enzyme activity** (100% to 60%) reduces efficiency of respiratory enzymes; slows breakdown of substrates in pathways such as **glycolysis** and **Krebs cycle**; reduces overall metabolic rate.*
*Reduced **muscle tissue activity** (100% to 55%) results from low ATP availability; **actin-myosin interactions** in muscle fibres are impaired; leads to reduced movement and slow growth.*
*Reduced **gill surface efficiency** (100% to 65%) lowers effective surface area for gas exchange; decreased diffusion of oxygen into blood reduces oxygen transport to tissues.*
*Low oxygen conditions force partial reliance on anaerobic metabolism; leads to accumulation of **lactic acid**, causing fatigue and reduced activity.*

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Cloudy water indicates high organic matter; microbial respiration further reduces oxygen levels, worsening hypoxic stress.

(b) Strategies

Reducing stocking density *decreases competition for oxygen; improves oxygen availability for each fish; enhances aerobic respiration and ATP production.*

Aeration of water *increases dissolved oxygen levels; supports efficient function of the electron transport chain and energy production.*

Introducing aquatic plants *increases oxygen production through photosynthesis; improves water quality and oxygen availability.*

Water circulation or replacement *removes waste products and increases oxygen diffusion; maintains optimal cellular respiration conditions.*

Monitoring water quality *ensures early detection of low oxygen levels; allows timely intervention to protect fish physiology.*

Adjusting feeding rates *prevents excess organic matter accumulation; reduces microbial respiration that consumes oxygen.*

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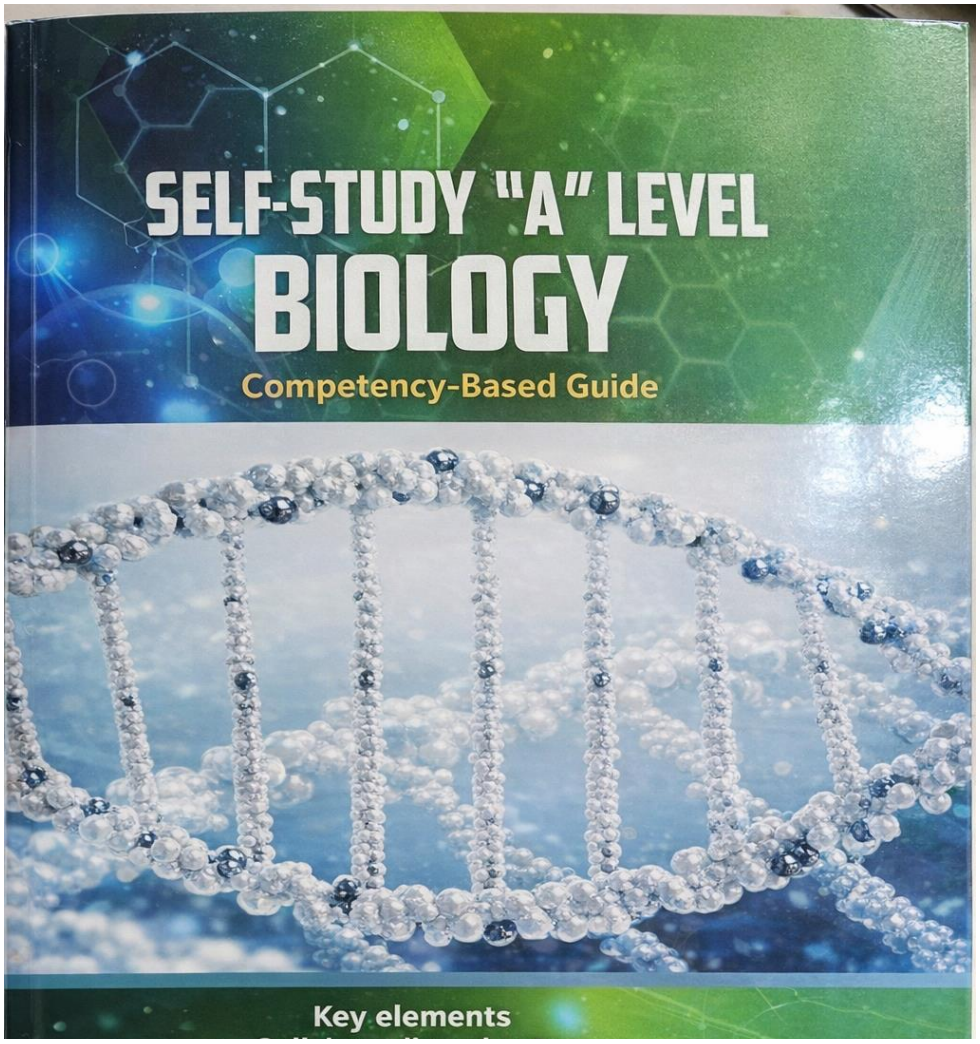
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